

Winter solar irradiation on roof surfaces

Bochum, North Rhine-Westphalia · Solar radiation analysis in ArcGIS Pro (Spatial Analyst)

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The task, its parameters and the data were set by the course, where the core workflow was demonstrated in a guided session. I repeated the analysis on my own; the data checks, the treatment of roof edges, the comparison with climate data, the map and this assessment are my own work.

1 Question

How much solar energy reaches each roof surface in the study area during the first ten days of the year, in watt-hours per square metre? The course sets the atmosphere: 40 % of the radiation arriving at the top of the atmosphere reaches the ground, and 40 % of that arrives indirectly (the course text says "by reflection"; physically it is mostly scattering). In the tool these become a transmissivity of 0.4 and a diffuse proportion of 0.4.

2 Data and source

Dataset	Content	Source
Surface model (DOM)	Buildings and vegetation included; 725 × 342 cells of 1.000 × 0.9996 m (resampled); heights 82–156 m; ETRS89 / UTM 32N	Course data; no metadata supplied
Orthophoto	RGB, same extent; used only as map background	Course data
Roof surfaces	62 polygons, geometry only; adjacent pairs indicate one polygon per roof plane	GIS-Akademie Berlin (exercise data)

The study area is the extent of the surface model, about 725 × 342 m in Bochum (51.51° N, 7.22° E). The 62 surfaces lie in its southern part; the allotment gardens to the north are not included, and the course does not say how the surfaces were chosen.

3 Method

- **Settings.** Extent, cell size and snap raster were set to the surface model before any raster tool. Its cells are not exactly 1 m, so "same as DOM" rather than a typed value.
- **Irradiation.** Raster Solar Radiation on the whole surface model, from 1 January 00:00 to 11 January 00:00 (UTC); uniform sky, diffuse proportion 0.4, transmissivity 0.4; slope and aspect derived by the tool from 3 × 3 cells. No mask: the shade of neighbouring trees and buildings has to enter the calculation. Outputs: total irradiation (kWh/m²) and direct-sun duration (h).
- **Roof edges.** At 1 m, the cells along a roof outline partly show walls and ground. The polygons were shrunk by 1 m (Pairwise Buffer, −1 m, no dissolve), so each mean rests on interior cells only.
- **Values per surface.** Zonal Statistics as Table (all statistics) on the shrunk polygons, with the original object ID as zone field; mean × 1,000 gives Wh/m². The same statistics on the original polygons serve as a comparison. Both were written to a copy of the roof layer with Join Field.
- **Reliability.** A surface without interior cells gets no value; one with fewer than 10 interior cells (10 m²) is flagged as low confidence.

Checks: the longest direct-sun duration, 77.8 h, is just below the roughly 80 h of daylight at 51.5° N over these ten days, so sun geometry and horizon behave as expected. AREA matches COUNT × 1 m². Warning 010566 (some zones not rasterized) matches the five shrunk polygons that contain no cell centre; a sixth vanished in the buffer. An intermediate copy of the roof layer had renumbered its object IDs, which shifted values between roofs from row 9 on; the join was rebuilt on a copy that keeps the IDs and checked against the zonal tables.

4 Result

Roof surfaces	Minimum	Median	Maximum
56 of 62 with a value	273 Wh/m ²	504 Wh/m ²	1,591 Wh/m ²

South-facing planes receive the most: up to 1,591 Wh/m², nearly six times the least favoured surface. The distribution is bimodal: 34 of the 56 surfaces receive less than about 600 Wh/m², 7 more than about 1,430, and few lie in between, so the median describes a typical surface better than the mean (729). The map shows the pattern: on buildings with an east–west ridge the south plane falls in the upper classes and the north plane in the lower ones, while planes facing east or west sit in between. With the sun at most about 16° above the horizon, the orientation of a plane decides more than where the building stands.

Edge cells matter for single surfaces, not for the overall picture. Against the original polygons, the shrunk ones change a surface's mean by a median of 6 % (mean 9 %), in both directions (median signed difference −0.1 %). Differences of 15–23 % occur on mid-sized planes too, and up to 78 % on the smallest. Six small roof sections (5.6–8.2 m²) have no value, and 8 surfaces rest on fewer than 10 cells, two on a single cell.

5 Analytical limits

- **The absolute values are far too low; the ranking is the result.** DWD's long-term January mean for Germany is 23 kWh/m² on a horizontal surface (1981–2020), roughly 5–7 kWh/m² for these ten days. The model gives at most 1.9 kWh/m² for any cell and 1.6 for the best roof plane, which faces south and in winter should receive more than a horizontal surface. In Esri's model, transmissivity refers to the vertical path and is raised to the relative path length, about 3.6 at a 16° sun: $0.4^{3.6} \approx 0.04$, so little direct radiation gets through. The course script gives 0.4 as a typical value for Germany and points to the Global Solar Atlas for more precise ones.
- **Ten January days are not a year.** With a low sun, orientation and shade dominate; over a year, flat and east- or west-facing roofs catch up. The values say nothing about annual yield.
- **Duration is geometry, not weather.** The 77.8 h are hours in which the sun stands above the local horizon; clouds enter only through the constant transmissivity.
- **The surface model sets the horizon.** Obstacles beyond its edge are missing, so planes near the southern edge may be overestimated: at this sun height a 10 m obstacle casts a shadow of about 35 m. The orthophoto shows summer foliage; if the model was captured at the same time, deciduous trees shade more in the model than in January.
- **Orientation was read from the map, not measured.** Aspect per plane was not extracted, so the explanation by orientation rests on the visual pattern.
- **Not modelled:** module efficiency, temperature, inverter losses, usable area (setbacks, chimneys, dormers) and roof statics. Irradiation is one input to a solar-potential assessment, not the yield.